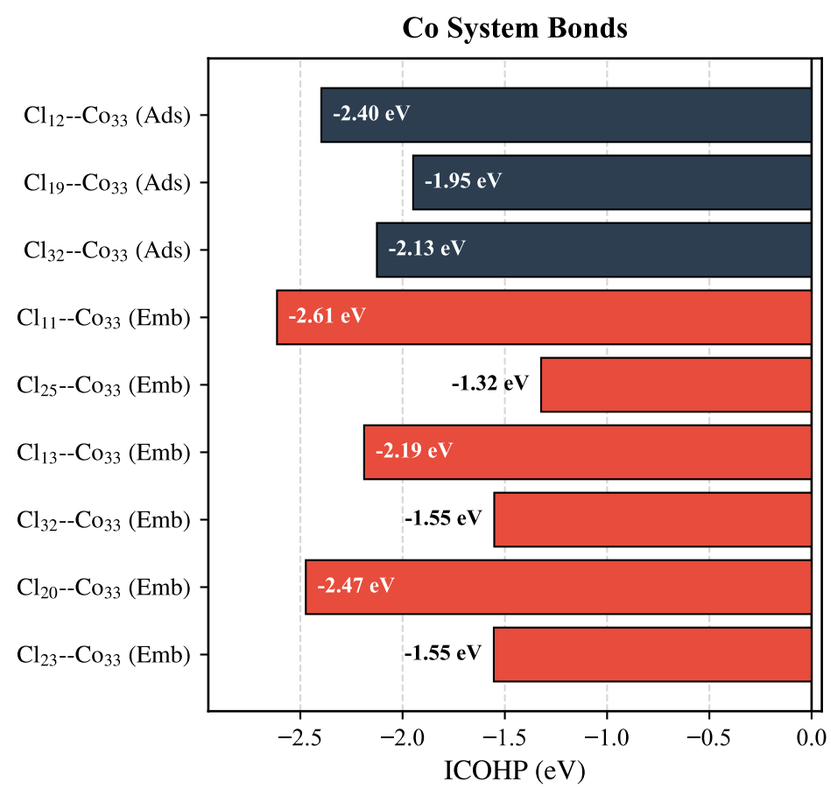
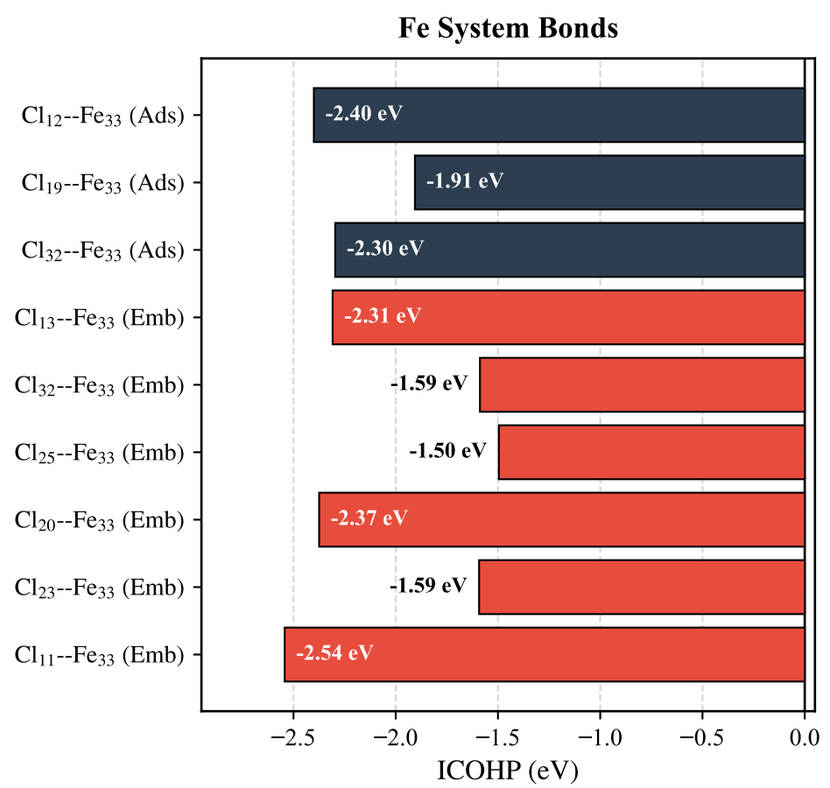
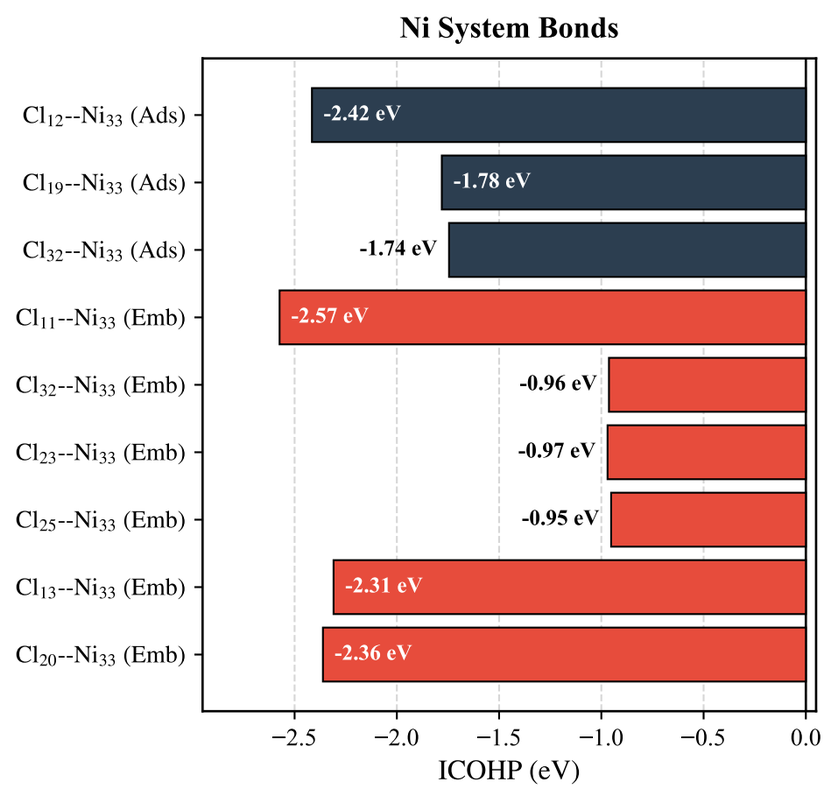
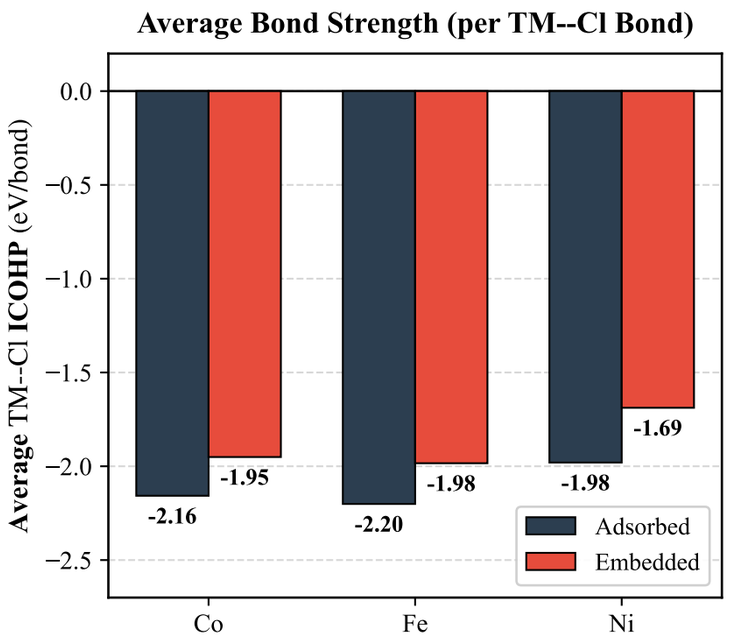
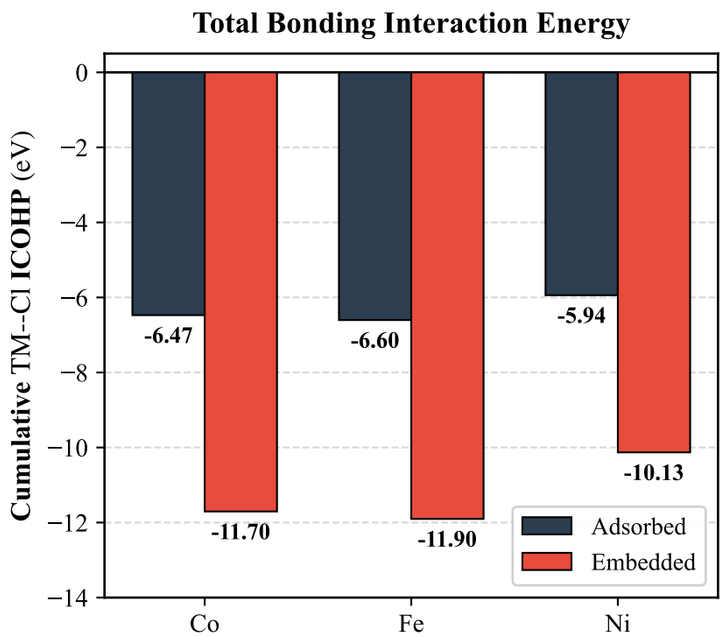

Co System Bonds
Cl12--Co33 (Ads)
Cl19--Co33 (Ads)
Cl32--Co33 (Ads)
Cl11--Co33 (Emb)
Cl25--Co33 (Emb)
Cl13--Co33 (Emb)
Cl32--Co33 (Emb)
Cl20--Co33 (Emb)
Cl23--Co33 (Emb)
-2.40 eV
-1.95 eV
-2.13 eV
-2.61 eV
-1.32 eV
-2.19 eV
-1.55 eV
-2.47 eV
-1.55 eV
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
ICOHP (eV)


Fe System Bonds
Cl12--Fe33 (Ads)
Cl19--Fe33 (Ads)
Cl32--Fe33 (Ads)
Cl13--Fe33 (Emb)
Cl32--Fe33 (Emb)
Cl25--Fe33 (Emb)
Cl20--Fe33 (Emb)
Cl23--Fe33 (Emb)
Cl11--Fe33 (Emb)
-2.40 eV
-1.91 eV
-2.30 eV
-2.31 eV
-1.59 eV
-1.50 eV
-2.37 eV
-1.59 eV
-2.54 eV
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
ICOHP (eV)


Ni System Bonds
Cl12--Ni33 (Ads)
Cl19--Ni33 (Ads)
Cl32--Ni33 (Ads)
Cl11--Ni33 (Emb)
Cl32--Ni33 (Emb)
Cl23--Ni33 (Emb)
Cl25--Ni33 (Emb)
Cl13--Ni33 (Emb)
Cl20--Ni33 (Emb)
-2.42 eV
-1.78 eV
-1.74 eV
-2.57 eV
-0.96 eV
-0.97 eV
-0.95 eV
-2.31 eV
-2.36 eV
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
ICOHP (eV)

(a) (b) (c)


Average Bond Strength (per TM--Cl Bond)
Average TM--Cl ICOHP (eV/bond)
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
-2.16
-1.95
-2.20
-1.98
-1.98
-1.69
Adsorbed
Embedded
Co
Fe
Ni


Total Bonding Interaction Energy
Cumulative TM--Cl ICOHP (eV)
0
−2
−4
−6
−8
−10
−12
−14
-6.47
-11.70
-6.60
-11.90
-5.94
-10.13
Adsorbed
Embedded
Co
Fe
Ni

(d) (e)